

Michael Shi

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Education

University of Alberta, Bsc in Computer Science with Minor in Business

09/2021 - 12/2026

- Awards: Undergraduate Leadership Award Scholarship, 4x Faculty of Science Scholarship, Dean's Honor Roll

Experience

Junior Software Developer (Part time), INVIDI Technologies - Edmonton, AB

05/2025 - Present

- Built and maintained Kotlin-based microservices supporting end-to-end ad campaign distribution, real-time subscriber targeting and segmentation, and integrations with external marketing and analytics platforms
- Led the team to restructure our AWS cloud infrastructure to reduce cost by implementing Intelligent-Tiering and Storage lifecycles for our S3 buckets. This resulted in a cost reduction of **\$18,000 (74.64% reduction)** for my team and my solutions was adopted as the blueprint for **600+** developers
- Caught inefficiencies in our CI/CD process, took initiative to automated our Bitbucket pipelines for all of our components. Created the infrastructure to deploy ECR images to EC2 instances and deploy Artifactory images to Kubernetes clusters, saving 3 hours of dev time every week as well as reducing rollout times by **60%**
- Led monitoring and logging infrastructure improvements by implementing Datadog metrics and centralized logging, reducing bug identification time by **80%** through automated alerting and detailed error traces
- Responsible for on-call incidents such as fixing production nodes failures, resolving critical bugs in Docker containers and REST APIs, this decreased our product downtime during out-of-office hours

Software Developer Intern, INVIDI Technologies - Edmonton, AB

05/2024 - 05/2025

- Migrated a our backend database from Apache Ignite to AWS Aurora PostgreSQL through schema redesign and refactoring, reducing our EC2 costs by **\$38,000 (89.26% reduction)** and eliminating false-positive on-call incidents
- Led an overhaul for our authentication design by splitting our OAuth permissions into SPA and M2M clients using TypeScript, improved authentication flow between our Vue.js frontend and Kotlin backend, speeding up the process by **30%**
- Assumed **Scrum Master** responsibilities during transition period; led all Agile ceremonies including sprint planning, daily standups, and retrospectives, ensuring continuous team productivity.

Undergraduate Teaching Assistant, University of Alberta - Edmonton, AB

09/2023 - Present

- Mentored over **1000** student in various Computer Science courses and using different programming languages
- Took on **Project Management** responsibilities for multiple software engineering teams, facilitated sprint planning while guiding teams through full-stack Android app development, ensuring best Agile and DevOps practices, resulting in **95%** of the teams receiving A and A+ for their projects
- Delivered interactive labs to students covering topics such as data structures, algorithms and automation in languages such as C, Bash, Java, and Typescript. Resulting in **20%** higher average scores compared to previous semesters
- Faced time-consuming and inconsistent grading processes, took initiative to automate grading using Python, reducing grading time by **90%** while ensuring consistent evaluation across both courses

Projects

StudyAmp: Nathacks 1st Place

github.com/NatRunners/StudyAmp

- Led a team of **5** to build a web-based attention monitoring solution using **EEG** tracking technology, securing **\$4,000** in seed funding to expand into a mobile app
- Engineered the full-stack system deployed on Vercel and Supabase, building scalable infrastructure to handle **10000+** concurrent users with low-latency performance during beta
- Developed Python backend for real-time brainwave processing, achieving **95%** accuracy in attention scoring through optimized data processing
- Built responsive frontend with Next.js and Framer Motion, creating smooth animations and polished UI that significantly boosted user engagement
- Caught a major pain point through beta feedback, users couldn't efficiently review long recordings, and quickly integrated Gemini AI to summarize audio at attention drops, cutting review time by **70%**

Technical Skills

Languages: Python, C, C++, Java, Kotlin, R, JavaScript, TypeScript, Bash

Frameworks: Node.js, REST API, Express.js, Flask, FastAPI, React, Next.js, Vue.js, Tailwind

Developer Tools: Git, Docker, Kubernetes, MongoDB, Firebase, PostgreSQL, AWS, Gradle, Terraform, Datadog, Snyk